

Lab 1.4: Downlink Handling

Course 1: LoRa LoRaWAN Fundamentals

USP Training Series

1 Lab Overview

Lab Objectives

By the end of this lab, you will be able to:

- Understand Downlink Handling concepts
- Implement working solution
- Test and verify functionality

Prerequisites

Required: Completion of previous labs

Hardware: Xiao nRF54L15 + LR1120 board

2 Lab Duration

Estimated Time: 60 minutes

3 Background

This lab focuses on Downlink Handling. Refer to Course 1 presentation for theoretical background.

4 Lab Instructions

4.1 Step 1: Setup

Configure your development environment for this lab.

4.2 Step 2: Implementation

Follow the solution code below to implement the required functionality.

4.3 Step 3: Build and Test

```
1 west build -b xiao_nrf54l15 .
2 west flash
```

5 Solution

Solution

Note: Complete solution code is available in the Lab_Solutions.md document, Section 1.4.

Key implementation points:

- Refer to Lab_Solutions.md for full working code
- All build configurations included
- Expected output documented
- Troubleshooting guide provided

See doc/training/Lab_Solutions.md for complete implementation details.

6 Expected Output

Refer to Lab_Solutions.md Section 1.4 for expected serial console output.

7 Deliverables

1. Working source code implementation
2. Serial console log showing functionality
3. Lab report with observations
4. Screenshots from network server (if applicable)

8 Troubleshooting

Common issues and solutions are documented in Lab_Solutions.md.